

Liquid Digital Twins Based on Magnetic Fluid Toys

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ABSTRACT

As a new type of functional material, magnetic fluid has both the fluidity of liquid and the magnetic properties of solid magnetic material. By controlling the magnets, one can simulate the effect of manipulating liquids like a sea emperor. This will provide new ideas for the multiverse of the metaverse. Not only that, magnetic fluids also have very important applications in astrophysics, controlled thermonuclear reactions and even the medical industry. Therefore, this paper hopes to provide a control idea for the future application of magnetic fluid by performing Digital Twins simulation of magnetic fluid.

Keywords: Magnetic Fluid; Digital Twins; Metaverse; Visualization;

Index Terms: Human-centered computing—Human computer interaction (HCI) —Interactive systems and tools—User interface management systems

1 INTRODUCTION

Magnetic fluid is a magnetic colloidal suspension formed by suspending a large number of magnetic micro-nano particles in water or oil. It can be traced back to the first patent for the preparation of magnetic fluid obtained by Papell in the United States in 1963. successfully used in the sealing of NASA aerospace products. Since then, people have started to develop and apply magnetic fluid, a new material, and have made new progress. So far, magnetic fluid has been used in mechanical engineering, electronic equipment and other industries for decades. In recent years, scientists have continued to find new application scenarios for it, such as cancer treatment, drug delivery in the body, oil spill treatment, etc. .

With the continuous deepening of the research on magnetic fluid, this also brings a new question: how to operate the magnetic fluid precisely to complete the precise operation? In this paper, the Digital Twins simulation of magnetic fluid is carried out, which provides a new idea for its precise operation.

2 THEORY

This paper models the magnetic fluid based on the Digital Twins. Starting with a simple magnetic fluid toy, a complete Digital Twins model of the magnetic fluid is established, and the simultaneous interpretation interaction of the Digital Twins is initially realized.

2.1 Magnetic Fluid

With the continuous in-depth research of magnetic fluids by scientists, magnetic fluids have been developed as liquid propellants from the beginning, and developed a number of functions including mechanical engineering, medical treatment and other fields.

Magnetic nanoparticle hyperthermia technology researched by Professor Neil Telling: Provide energy to magnetic nanoparticles, and the part of the energy that it cannot store is released as heat,

which can heat nearby cells. Studies have shown that cancer cells are more heat-labile than normal cells, and it is only necessary to use this technique to heat the cells to 40°C to 42°C to kill cancer cells. By remotely heating magnetic nanoparticles, it is expected to destroy tumors deep in the body.

MagForce has developed Nanotherm, a magnetic fluid product for the treatment of brain tumors, which was approved for use in the European Union in 2011. During the treatment, the patient lies in a machine that generates an alternating magnetic field, which emits a focused magnetic field that acts on the tumor site where the magnetic fluid has been injected to heat the cells. They are currently conducting clinical trials for prostate cancer treatment in the United States and announced in February 2021 that they have seen efficacy in 10 patients.

This ferrofluidic heating technology isn't limited to cancer treatments, as Professor Telling's team used it for the first time in 2019 to kill the parasite that causes the tropical disease leishmaniasis. In addition, researchers at Kent State University in Ohio recently used the technique to remove the amyloid plaques associated with Alzheimer's disease in human brain cells.

Figure 1 demonstrates the capabilities of magnetic fluids as medical robots.[1]

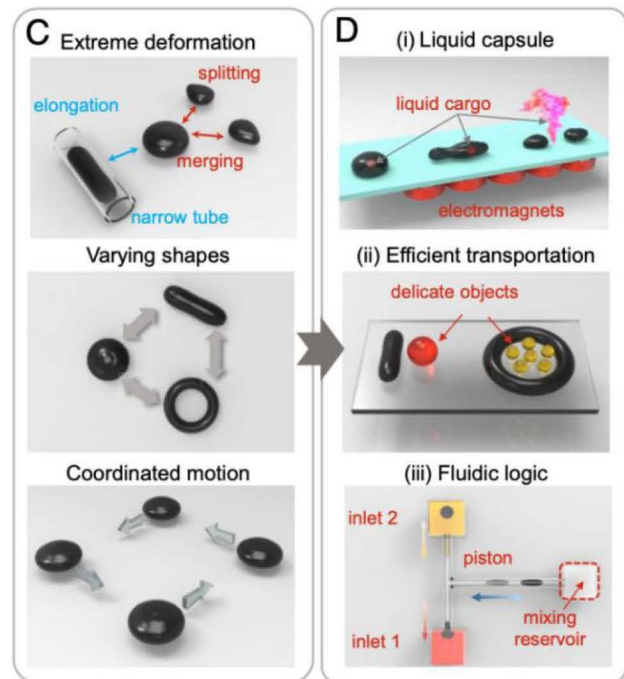


Figure 1: The three pictures on the left show the ability of the droplet to deform and move; the three pictures on the right show the functions of the droplet to carry goods.

2.2 Digital Twins

The conceptual model of the Digital Twins first appeared in 2003, by Grieves M. W. The professor proposed it in the product lifecycle management (PLM) course of the University of

Michigan, which was called the "mirror space model" at that time [2], and was later defined as the "information mirror model" in the literature [3]. and "Digital Twins". In 2010, the National Aeronautics and Space Administration (NASA) introduced the concept of Digital Twins for the first time in the space technology roadmap [4].

So far, Digital Twins have been used in many medical applications. Hewlett-Packard and the Federal Institute of Technology Lausanne (EPFL) have cooperated in the "Blue Brain" project, using supercomputers to create a digital model of the brain; Siemens has established A "Digital Twins" model, Intel, Bayer, Hewlett Packard Enterprise, and Pfizer are all working on a Digital Twins heart model [5]. France-based Sim&Cure applies Digital Twins technology to the personalized treatment of intracranial aneurysms. Sim&Cure strives to avoid preventable medical errors, helps neurological health professionals choose implant sizes to fit the aneurysm perfectly, and allows clinicians to build patient-specific Digital Twins of aneurysms and arterial trees from their rotational angiography scans mold. Experts can then simulate the implant deployment process and obtain a precise visualization of the interaction between the aneurysm and the implant.

3 DISCUSSION

Although there are few papers related to liquid digital twins, more and more people are starting to pay attention to liquid digital twins. On December 23, 2021, the Ministry of Water Resources of China held a working conference on promoting the construction of digital twins watersheds, emphasizing the importance of digital twins watersheds, which is also a digital twins of liquids. It can be seen that in the future, more and more people will start to pay attention to the digital twins of liquids. Therefore, our project is of very practical significance for research.

3.1 Digital Twins Modeling of Magnetic Fluids

We built a simple model of the droplet state of a ferrofluid using Unity, as shown in Figure 2.

Our preliminary design synchronizes the droplet model with the droplet in the ferrofluidic toy, so that our droplet model can perform the same action as the ferrofluidic droplet. However, at present, we can only synchronize the droplets in similar positions, and cannot make the droplets achieve the effect of "blooming" of the magnetic fluid. The synchronization effect will be shown in the video material.

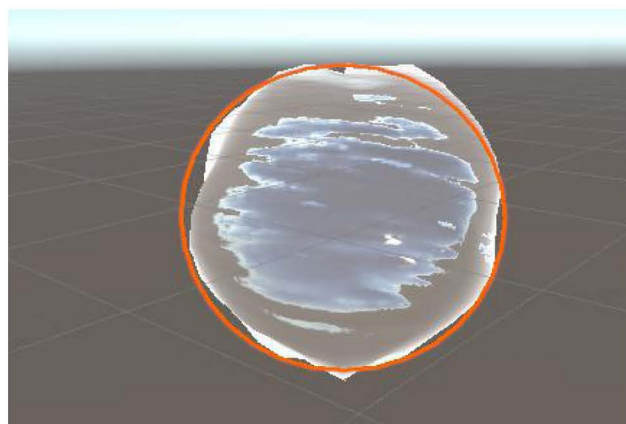


Figure 2: Magnetic fluid droplet model

3.2 Simulation of liquid trajectory

Our simulation of liquid trajectory is also realized through unity, because as long as we set the appropriate shader parameters, we can get a very beautiful liquid trajectory.

However, in the simulation process, we also found some problems to be improved. First of all, we can control the trajectory of the liquid, but we can't achieve accurate control. For example, when the magnetic fluid is attracted by the magnetic force, it will bloom, which is difficult to simulate. Therefore, we will make some improvements in the future. Realize liquid digital twins.

In addition, there is a problem to be solved in liquid digital twin, that is, how to use the sensor to perfectly capture the real-time state of liquid, and then transmit it to the computer to realize the simultaneous interpretation of liquid digital twin.

Figure 3 shows the trajectory of the liquid.

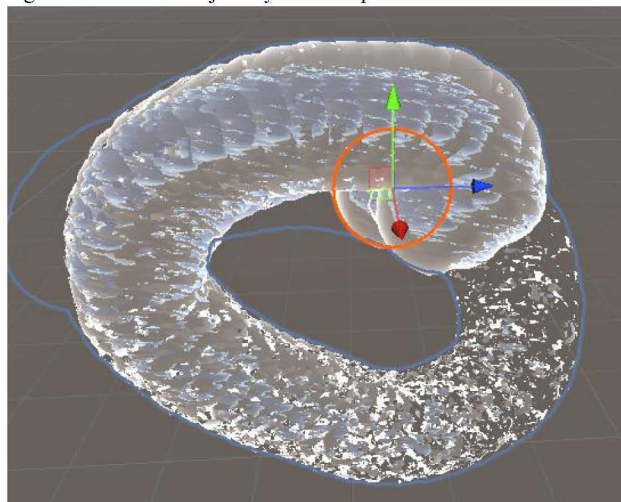


Figure 3: the trajectory of the liquid.

4 CONCLUSION

To a certain extent, the metaverse is the Digital Twins of everything, and the most attractive thing about the metaverse is the ability to experience different digital roles. If we get a liquid digital twins and can control it, then we may have hope in the future in the metaverse. Incarnate in the universe to become a sea king and control the water body. Furthermore, we can use the digital twins of liquid to detect seawater in real time, detect water level, and apply it to the fields of water resources management and fishery. Hope it can provide a reference for future research work.

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